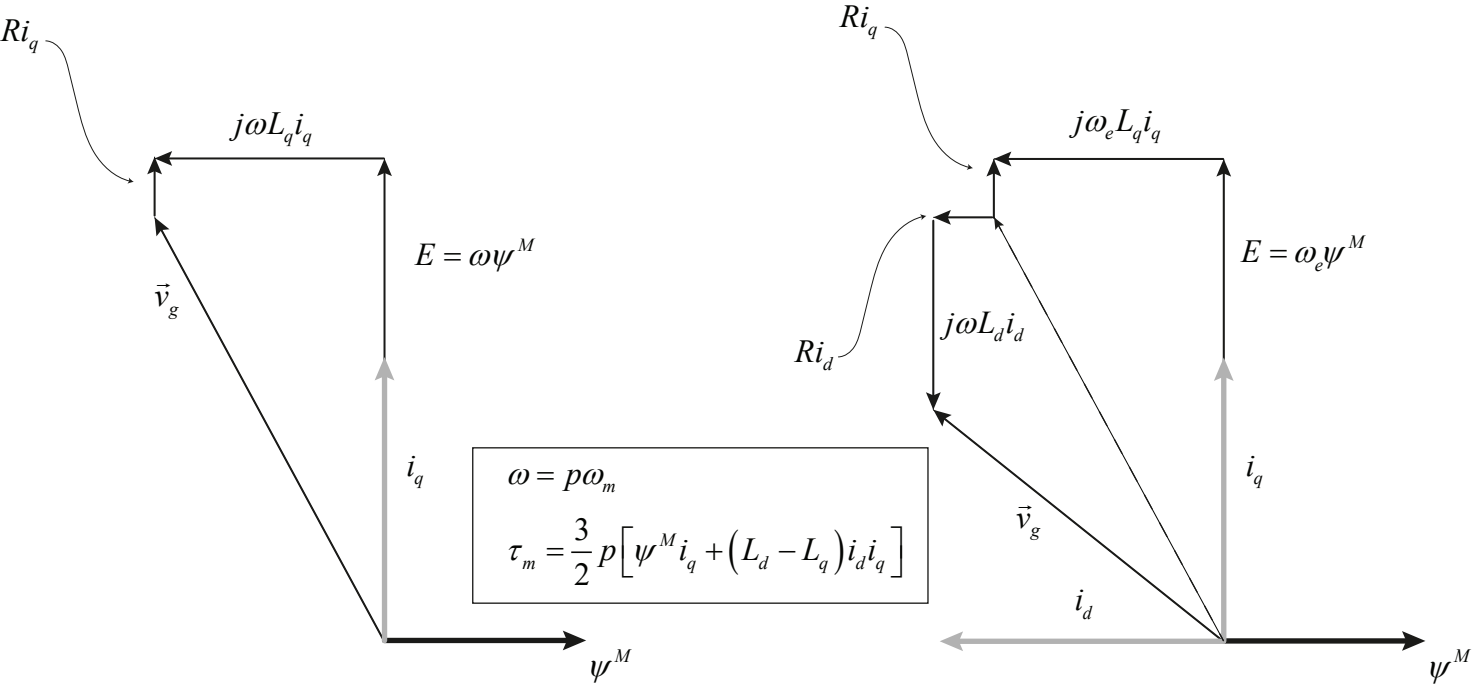
Ri_q
$j\omega L_q i_q$
$E = \omega\psi^M$
$\vec{v}_g$
i_q
ψ^M
$\omega = p\omega_m$
$\tau_m = \frac{3}{2} p\left[\psi^M i_q + \left(L_d - L_q\right) i_d i_q\right]$
Ri_q
$j\omega_e L_q i_q$
$E = \omega_e \psi^M$
Ri_d
$j\omega L_d i_d$
i_q
$\vec{v}_g$
i_d
ψ^M